

Contents

S.No.	Topic Page	Page No.
1.	Introduction	1
2.	Bureau of Indian Standards	4
3.	Definitions	7
4.	Raw Materials	9
5.	Capacitor Standards	13
6.	Testing Requirements	26
7.	Capacitor Selection	28
8.	National Electrical Code of India	31
9.	Regulatory Requirements	34
10.	References	36
	Annex 1 List of Standards	38



CHAPTER I

INTRODUCTION

CHAPTER I

INTRODUCTION

Power capacitors, fundamental components of electrical systems, have a history deeply intertwined with the development of power distribution technologies. As alternating current (AC) systems gained prominence in the early 20th century, the issue of reactive power emerged as a challenge in power transmission and distribution. Inductive loads, including motors and transformers, drew reactive power from the grid, leading to a lagging power factor and inefficient energy consumption. To mitigate this, power capacitors were developed to provide reactive power compensation.

The earliest power capacitors were simple fixed capacitors manually switched in or out of circuits to correct power factor. These capacitors played a crucial role in improving the efficiency of electrical systems, but their manual operation limited their effectiveness in dynamically changing load conditions. However, with the advancement of technology, automatic capacitor banks (ACBs) were introduced. Equipped with sophisticated controllers, ACBs could dynamically adjust capacitor switching based on real-time measurements of power factor, enabling more precise and efficient power factor correction.

The evolution of power capacitors has significantly enhanced the efficiency and reliability of electrical networks. By providing reactive power compensation, power capacitors optimize power factor, reducing transmission losses and improving energy efficiency. Moreover, they stabilize voltage levels, enhance system performance, and contribute to the overall stability of electrical grids.

In modern electrical systems, power capacitors continue to play a vital role in optimizing power quality and ensuring efficient energy usage. They are deployed in various applications, including industrial plants, commercial buildings, and utility-scale power generation facilities. Furthermore, advancements in capacitor technology, such as the development of specialized capacitor materials and innovative capacitor bank designs, continue to drive improvements in power system performance and reliability.

1.1 Importance of Power Capacitors in Power Systems

- **Power Factor Correction:** Power capacitors are primarily used to correct power factor. Power factor refers to the ratio of real power (used to perform work) to apparent power (delivered by the utility). A low power factor signifies inefficient power utilization and leads to increased energy costs, higher transmission losses, and reduced equipment capacity. Power capacitors inject leading reactive power into the system, counteracting the lagging reactive power consumed by inductive loads (motors, transformers etc) and improving the overall power factor.
- **Voltage Regulation:** Power capacitors help maintain voltage stability within the power system. During peak demand periods, voltage levels can drop due to increased current flow. Capacitors act as a reservoir of reactive power, releasing it back into the system to support voltage and prevent voltage sags that can damage sensitive equipment.

- **Improved System Efficiency:** By correcting power factor and regulating voltage, power capacitors contribute to a more efficient power system. Reduced losses in transmission lines translate to lower energy consumption and cost savings for both utilities and consumers.
- **Smoothing DC Voltage:** In power converters that convert AC to DC (rectifiers), capacitors help smooth out the pulsating DC output voltage, providing a more stable and continuous supply for electronic devices.
- **Filtering Harmonics:** Power electronic converters can introduce harmonic distortions into the power system. Power capacitors, in conjunction with filters, can help mitigate these harmonics, ensuring cleaner power and protecting sensitive equipment from damage.
- **Energy Storage:** In emerging applications like renewable energy integration, power capacitors can be used for short-term energy storage. They can store excess energy generated from solar or wind sources and release it back into the grid when needed.

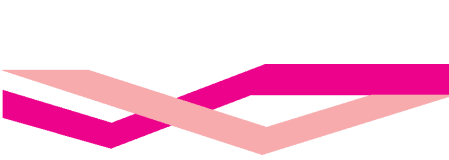
In essence, power capacitors play a crucial role in promoting a more sustainable and environmentally conscious power system for the future.

1.2 Need for Standardization

With the widespread use of power capacitors in various applications, standardization becomes crucial for several reasons:

- **Interoperability:** Standardized capacitors ensure compatibility with different system components and facilitate easy integration into existing power systems.
- **Safety and Reliability:** Standards define safety requirements for capacitor design, testing, and installation, minimizing the risk of failures and ensuring reliable operation.
- **Performance Consistency:** Standardization guarantees a certain level of performance consistency across different capacitor manufacturers, allowing engineers to make informed choices based on well-defined specifications.
- **Fair Competition:** Standardized capacitors create a level playing field for manufacturers, promoting fair competition and innovation within the industry.

This handbook aims to provide a foundational understanding of standardisation happening in India in the field of power capacitors. However, this does not intend to replace the Indian Standards. It is always advisable to refer the latest edition of Indian Standards available from the portal of Bureau of Indian Standards. All the indigenous standards are available for free download after registering in the portal but the adopted International standards are paid publications.



CHAPTER II

BUREAU OF INDIAN STANDARDS

CHAPTER II

BUREAU OF INDIAN STANDARDS

Bureau of Indian Standards (BIS), the National Standards Body (NSB) of India was established under the BIS Act, 1986 and came into existence on 1 April 1987 assuming the functions of the erstwhile Indian Standards Institution (ISI). The ISI came into being on 6 January 1947. The BIS Act, 2016 came into force on 12 October 2017 superseding BIS Act 1986. BIS Act, 2016 provides for the establishment of a national standards body for the harmonious development of the activities of standardization, conformity assessment and quality assurance of goods, articles, processes, systems and services and for matters connected therewith or incidental thereto.

BIS through its core activities of standardization and conformity assessment, has been benefiting the national economy by providing safe, reliable and quality goods; minimizing health hazards to consumers; protecting the environment, promoting exports and imports substitutes; controlling over proliferation of varieties, etc. The standards and certification schemes of BIS apart from benefitting the consumers and industry also support various public policies especially in areas of product safety, consumer protection, food safety, environment protection, building and infrastructure development, etc.

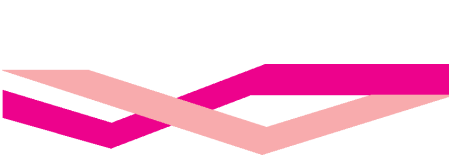
BIS also represents India in International Standards bodies like International Organization for Standardization (ISO) and International Electrotechnical Commission (IEC) and participates actively in the international standardization work undertaken in these bodies. BIS presents the national viewpoints on new areas taken up for international standardization and on various draft international standards during the process of development of these standards so that the country's interest is protected and reflected in these standards. This also enables the BIS technical committees to consider adoption of the International Standards as Indian Standards, with or without modifications, in order to enable our products and services to integrate with global trade and commerce.

Standards are as good as the people who formulate them. If a standard has to truly reflect the collective knowledge of the society and provide a reference point to all the stakeholders, it is important the more and more people contribute towards development of national and international standards.

2.1 Indian Standards on Power Capacitors

Power Capacitors Sectional Committee ETD 29 under the Electro Technical Division Council of BIS is responsible for standardization in capacitors. The standards specify uniform rules regarding performance, testing and rating, specific safety requirements and provide a guide for proper installation and operation. It is important that the capacitors manufactured and used shall comply with the relevant national standard.

Many of these standards are harmonized with the corresponding International Standard published by International Electrotechnical Commission (IEC) and are indicated by dual numbering in the fashion IS XXXXX /IEC YYYYY, or by single numbering in the fashion IS/IEC XXXXX.



Some of the important Indian Standards published by this committee and few related standards published by other committees are as given in Annexure 1. Whether they are harmonized with international standards or not is also specified.

2.2 Harmonization

Agreement on Technical Barriers to Trade under WTO encourage member countries to reduce technical barriers by using international standards in their technical regulations. Aligning our standards with International standards also empower our industry to be globally competitive. However, it is important the national interests are protected and distinct national conditions are taken care of in our standards. This also points to the importance of our participation in the international standardisation activities of IEC, ISO etc so that our interests are incorporated in the international standards formulated.



CHAPTER III

DEFINITIONS

CHAPTER III

DEFINITIONS

Bureau of Indian Standards has published a series of standards under IS 1885 on Electrotechnical Vocabulary. The series has got about 50 standards covering various aspects of electrical and electronics field.

There are two standards relevant to power capacitors in this series:

IS 1885 (Part 42) : 1993 Electrotechnical vocabulary: Part 42 Power Capacitors

IS 1885 (Part 45) : 1977 Electrotechnical vocabulary: Part 45 capacitors

A glimpse of few definitions are given below:

- Capacitor element – A device consisting essentially of two electrodes separated by a dielectric material, its intended characteristic being capacitance.
- (capacitor) unit - An assembly of one or more capacitor elements in the same container with terminals brought out.
- (capacitor) bank - A number of capacitor units connected so as to act together.
- Capacitance (Quantity) – The quantity of charge on one of the electrodes of a capacitor divided by the corresponding potential difference between the two electrodes, the influence of any other conductor being negligible.
- Rated output of a capacitor - The reactive power for which the capacitor has been designed. (It may be noted that the power capacitors are rated and designated by their kVAR output rather than their Capacitance for convenience.).
- Tangent of the loss angle (of a capacitor) - $\tan \delta$ (Delta) (abbreviation) - The ratio between the equivalent series resistance and the capacitive reactance of the capacitor at specified sinusoidal alternating voltage and frequency.

Electrotechnical Vocabulary is quick reference guide for any terms related to the electrical field.



CHAPTER IV

RAW MATERIALS

CHAPTER IV

RAW MATERIALS

4.1. Polypropylene (PP)

Polypropylene is one of the most widely used dielectric materials in power capacitors due to its excellent dielectric properties, high breakdown voltage, low dissipation factor, and good thermal stability. PP capacitors offer high energy storage density and long-term reliability, making them suitable for various power factor correction and harmonic filtering applications.

IS 11298 (Part 3/Sec 1) covers the requirements of plain and hazy polypropylene films (non-metallized) for capacitors. The material shall be made from polypropylene homopolymer and shall conform to IS 11298 (Part 1) : 1985. The standard specifies following requirements for the film:

Mechanical Properties :

- Thickness in microns
- Space factor for hazy film
- Width in mm
- Roll outside diameter
- Density
- Elongation at break
- Heat Shrinkage

Electrical Properties:

- DC Dielectric Strength
- Volume Resistivity
- Dissipation Factor – is a measure of the resistive loss caused by the dielectric
- Dielectric Constant – The specified value is 2.2 ± 0.05 . It determines the capacitance by the formula

$$C = K\epsilon_0 \frac{A}{d}$$

Where

C is the Capacitance,

K is Dielectric Constant of the material,

- $\epsilon_0 = 8.854 \times 10^{-12}$ farads per meter, Permittivity of Vacuum

A is the area of the plates, and
d is the thickness of the dielectric

(The formula will change suitably for cylindrical design)

- Electrical Weakspots

IS 11298 (Part 2) specifies the detailed and precise test method for each of the above tests.

4.2 Metallized Polypropylene Film

Metallized polypropylene film capacitors combine the dielectric properties of polypropylene with the high conductivity of metal coatings. The metal coating act as the conducting plate of the capacitor and polypropylene as dielectric. Zinc, Aluminium and Zinc-Aluminium alloy are specified as metals used for metalizing. Metallized film capacitors offer improved self-healing properties, reduced size, and enhanced reliability compared to traditional film capacitors with separate foil electrodes. They are widely used in power factor correction, motor run capacitors, and DC-link applications.

IS 11298 (Part 3/Sec 2) specifies requirements for Metallized Polypropylene film (MPP). The base polypropylene film shall confirm with the requirements of IS 11298 (Part 3/Sec 1) as given above and this standard, in addition specifies the following requirements:

Mechanical Tests-

- Thickness
- Width
- Length / Diameter
- Shrinkage
- Coating Adhesion
- Tram lines
- Tightness of winding

Electrical Tests –

- Resistance of metal layer
- Voltage proof test
- Power Factor
- Electrostatic charge.

Test methods are again as given in IS 11298 (Part 2).

4.3 Paper

Paper used as dielectric shall confirm to the following standards:

- IS 9335 (Part 3/Sec 2) : 1985 Specification for cellulosic papers for electrical purposes: Part 3 specifications for individual materials: Sec 2 capacitor paper

- IS 9335 (Part 3/Sec 4) : 1984 Specification for cellulosic papers for electrical purposes: Part 3 specifications for individual materials: Sec 4 electrolytic capacitor paper
- IS 9335 (Part 1) : 2024 Cellulosic Papers for Electrical Purposes Part 1 Definitions and General Requirements
- IS 9335 (Part 2) : 2024 Cellulosic Papers for Electrical Purposes Part 2 Methods of Test

Tensile strength, Moisture Content, Ash Content, Apparent density, pH, Electric Strength, Dissipation factor etc are some of the important quality parameters specified in the standards along with test methods for the same.

4.4 Impregnating Agents

Wound capacitor elements are often impregnated with mineral / synthetic oil or other impregnants which will flush out the air and moisture, and improve the dielectric strength of the capacitor. IS 13067:1991 gives the requirement of impregnants for use in power capacitor. This standard covers Ester Compounds and Synthesised Hydrocarbon Compounds. The standard specifies the following requirements:

- Colour
- Specific Gravity
- Refractive Index
- Coefficient of Thermal Expansion
- Neutralization Value
- Ash Content
- Water Content
- Kinematic Viscosity
- Pour Point
- Flash Point
- Chlorine Content
- Specific Heat
- Fire Point
- Dielectric Constant
- Dielectric Dissipation Factor
- Volume Resistivity
- Dielectric Strength

Test methods for each of the above tests are given in different cross referred standards.



CHAPTER V

CAPACITOR STANDARDS

CHAPTER V

CAPACITOR STANDARDS

5.1 Self-Healing Capacitors

Construction:

- These capacitors typically utilize metallized polypropylene films (MPP).
- The film dielectric is much thinner compared to non-self-healing types, allowing for higher capacitance density.
- These capacitors exhibit a **self-healing** property by causing vaporization of the metal upon encountering a localized high temperature caused by a small puncture or breakdown.
- The vaporization process effectively isolates the fault area, preventing further damage and allowing the capacitor to continue functioning.

Merits:

- **High Reliability:** The self-healing mechanism helps prevent complete capacitor failure due to minor internal breakdowns.
- **Extended Lifespan:** By recovering from minor faults, self-healing capacitors can offer a longer operational life compared to non-self-healing types.
- **Improved Safety:** The self-healing process helps contain internal faults, reducing the risk of catastrophic failures like explosions or fire.

IS 13340: SHUNT POWER CAPACITORS OF THE SELF - HEALING TYPE FOR AC SYSTEMS HAVING A RATED VOLTAGE UP TO AND INCLUDING 1000 V specifies the requirements of such capacitors.

This standard is published in two parts:

Part 1 General - Performance, testing and rating - Safety requirements - Guide for installation and operation

Part 2 Ageing test, Self - Healing test and Destruction test

- a) **Scope:** The standard is applicable to both capacitor units and capacitor banks intended to be used, particularly, for power-factor correction of a.c. power systems having a rated voltage up to and including 1 000 V and frequencies of 15 Hz to 60 Hz.
- b) **Specifications:** The standard specifies various parameters and characteristics of shunt capacitors, including:
 - Electrical properties such as rated voltage, frequency, and reactive power.
 - Performance criteria concerning temperature rise, losses, and durability.

- Overload conditions
- Safety requirements
-  Marking requirements
- Self-Healing capacitors are marked as “SH” or by the symbol
-  Internal fuses if present are indicated by the symbol
- Service conditions
- The standard defines different ambient air temperature categories in which the capacitor is designed to operate. The preferred values for lowest temperature are +5 °C, –5 °C, –25 °C, –40 °C, –50 °C. The highest values are represented by symbols A, B, C, or D with maximum of 40, 45, 50 or 55 °C.

The temperature category is marked as, for example, “–40/A”.

- Guide for installation and operation
 - Additional requirements for power filter capacitors
 - Formulae for information
- c) Testing:** The standard outlines testing procedures and methodologies to evaluate the performance and reliability of shunt capacitors.

The tests are generally classified into Routine tests and Type tests.

Routine tests – Carried out on each individual item during or after manufacture:

- Capacitance measurement and output calculation
Power capacitors are generally rated in terms of kVAR output which is calculated from the capacitance measured, based on the configuration.
- Measurement of the tangent of the loss angle ($\tan \delta$) of the capacitor –
Real capacitors have some internal resistance and dielectric losses. $\tan \delta$, also known as dissipation factor or loss tangent, quantifies the resistive loss in a capacitor. A lower $\tan \delta$ indicates a higher quality capacitor with less energy loss.

$$\tan \delta = \frac{ESR}{X_c}$$

where:

ESR = Equivalent Series Resistance (internal resistance of the capacitor)

$$X_c = \text{Capacitive Reactance} = \frac{1}{2\pi fC}$$

$\tan \delta$ meters measure impedances of a bridge circuit and calculate the loss angle.

- Voltage test between terminals, Voltage test between terminals and container – High voltage withstand tests for electrical safety.
- Test of the internal discharge device – Capacitors may have an internal discharge device to discharge the capacitor for safe handling, after disconnected from supply. Functioning of this device is checked through this test.
- Sealing test – Applicable when there is liquid dielectric in the capacitor.

Type tests – Tests done to confirm whether a particular design conform to the requirements of the standard:

- Thermal stability test and Measurement of the $\tan \delta$ of the capacitor at elevated temperature
To determine the stability of the capacitance during continuous operation at maximum rated ambient temperature conditions.
- Voltage test between terminals, Voltage test between terminals and container
- Lightning impulse voltage test between terminals and container
Capacitors intended for exposed installations shall not fail during lightning impulses.
- Discharge test
- Ageing test
Accelerated ageing test by subjecting capacitors to a series of charge-discharge cycles which shall not break down the capacitor and shall not cause variation in capacitance beyond a specified limit.
- Self-healing test
To check the self-healing break downs of capacitors and stability of capacitance during this process.
- Destruction test
Capacitor is applied with a high DC voltage to promote failures and electrical and mechanical safety is verified.

5.2 Non-Self-Healing Capacitors

Construction:

- These capacitors can utilize various dielectrics, including **film (with thicker films than self-healing types), electrolytic (Aluminium or tantalum), or ceramic.**
- The construction is simpler and often involves a thicker dielectric layer compared to self-healing capacitors.
- They do not incorporate any specific self-healing mechanism.

Merits:

- **Cost-Effective:** Non-self-healing capacitors are generally less expensive than self-healing types due to the simpler construction.
- **Compact Size:** They can achieve higher capacitance density for a given size compared to self-healing types due to the thicker dielectric layer.
- **Wide Range of Applications:** Available in various dielectric materials, non-self-healing capacitors cater to a broader range of applications with diverse voltage and temperature requirements.

IS 13585: SHUNT POWER CAPACITORS OF THE NON SELF - HEALING TYPE FOR AC SYSTEMS HAVING A RATED VOLTAGE UP TO AND INCLUDING 1000 V specifies the requirements.

This standard is published in three parts:

Part 1 General - Performance, testing and rating - Safety requirements - Guide for installation and operation

Part 2 Ageing test, Self - Healing test and Destruction test

Part 3 Internal Fuses

- a) Scope:** The standard is applicable to both capacitor units and capacitor banks intended to be used, particularly, for power-factor correction of a.c. power systems having a rated voltage up to and including 1 000 V and frequencies of 15 Hz to 60 Hz.
- b) Specifications:** The standard specifies various parameters and characteristics of shunt capacitors, including:
 - Electrical properties such as rated voltage, capacitance, and reactive power.
 - Performance criteria concerning temperature rise, losses, and durability.
 - Other parameters as discussed in the previous standard.
- c) Testing :** Testing details are same as that of Self – Healing capacitors, except that the self-healing test is not applicable.
- d) Safety Requirements:** The standard includes safety considerations to mitigate risks associated with the operation and maintenance of shunt capacitors. This involve guidelines for installation, discharge resistors, overpressure protection, and safety markings.

5.3 Shunt Capacitors for power systems above 1000 V

IS 13925: SHUNT POWER CAPACITORS FOR AC SYSTEMS HAVING A RATED VOLTAGE ABOVE 1000 V

This standard is published in four parts:

Part 1 General - Performance, testing and rating - Safety requirements - Guide for installation and operation

Part 2 Endurance Testing

Part 3 Protection of Shunt Capacitors and Shunt Capacitor Banks

Part 4 Internal Fuses

- a) **Scope:** The standard is applicable to both capacitor units and capacitor banks intended to be used, particularly, for power-factor correction of a.c. power systems having a rated voltage above 1 000 V and frequencies of 15 Hz to 60 Hz.
- b) **Specifications:** The standard specifies various parameters and characteristics of shunt capacitors, including:
- Electrical properties such as rated voltage, capacitance, and reactive power.
 - Mechanical aspects like dimensions, mounting arrangements, and environmental conditions.
 - Performance criteria concerning overload, temperature rise, losses, and durability.
 - Service conditions.
 - Marking Requirements
 - Guide for installation and operation.
- c) **Testing :**

Routine tests

- a) Capacitance measurement
- b) Measurement of the tangent of the loss angle ($\tan \delta$) of the capacitor
- c) Voltage test between terminals, AC voltage test between terminals and container
- e) Test of internal discharge device
- f) Sealing test
- g) Discharge test on internal fuses

Type tests

- a) Thermal stability test and Measurement of $\tan \delta$ at elevated temperature
- c) AC voltage test between terminals and container
- d) Lightning impulse voltage test between terminals and container
- e) Short-circuit discharge test
- f) Test of an external fuse in combination with a capacitor
- g) Disconnecting test on internal fuses.

Special Test - Endurance test - As this is a time consuming and expensive test, not included in the type tests but carried out in case required.

5.4 Series Capacitors

While generally Capacitors are connected in parallel across the load for power factor compensation, they are also connected in series in transmission lines for voltage stabilisation.

These capacitors shall confirm to IS/IEC 60143: SERIES CAPACITORS FOR POWER SYSTEMS

This standard is published in four parts:

Part 1 General

Part 2 Protective Equipment for Series Capacitor Banks

Part 3 Internal Fuses

Part 4 Thyristor Controlled Series Capacitors

- a) **Scope:** The standard is applicable to both capacitor units and capacitor banks intended to be used connected in series with an a.c. transmission or distribution line or circuit forming part of an a.c. power system having a frequency of 15 Hz to 60 Hz. The standard is primarily focussed on transmission application.
- b) **Specifications:** The standard specifies various parameters and characteristics of series capacitors, including:
 - Electrical properties such as rated voltage, capacitance, frequency range and reactive power.
 - Service conditions.
 - Performance criteria concerning temperature rise, losses, and durability.
 - Over Voltage Protector types
 - Insulation levels
 - Marking requirements.
 - Requirements for external fuses.
- c) **Testing:** The standard outlines testing procedures and methodologies to evaluate the performance and reliability of series capacitors. These tests includes:

Routine tests

- a) Capacitance measurement
- b) Capacitor loss measurement
- c) Voltage test between terminals
- d) AC voltage test between terminals and container
- e) Test on internal discharge device
- f) Sealing test
- g) Discharge test of internal fuses

Type tests

- a) Thermal stability test
- b) AC voltage test between terminals and container
- c) Lightning impulse voltage test between terminals and container
- d) Cold duty test

This test demonstrates the ability of the capacitor to withstand an overvoltage at the protective level followed by an overcurrent condition with the capacitors initially at the lowest ambient temperature.

- e) Discharge current test
- Special test – Ageing test

- d) **Safety Requirements:** The standard includes safety considerations to mitigate risks associated with the operation and maintenance of series capacitors. This involves guidelines for installation and operation, discharge resistors, container connection and protection of the environment.

5.5 Capacitors for Power Electronics

These are used for filtering and smoothing the voltage. They can be ac or dc, and self-healing or non self-healing.

IS/IEC 61071: CAPACITORS FOR POWER ELECTRONICS

- a) **Scope:** This standard applies to capacitors for power electronics applications having operating frequency of the systems in which these capacitors are used usually up to 15 kHz. This standard covers wide range of capacitor technologies for numerous applications, e.g. overvoltage protection, DC and filtering, switching circuits, energy storage, auxiliary inverters, etc
- b) **Specifications:** The standard specifies various parameters and characteristics of capacitors for use in power electronics, including:
 - Electrical properties such as rated voltage, capacitance, frequency range, etc
 - Service conditions
 - Operating temperature maximum 45, 55, 70 or 85 °C
 - Overloads.
 - Performance criteria concerning temperature rise, losses, and durability.
 - Marking requirements.
- c) **Testing:** The standard outlines testing procedures and methodologies to evaluate the performance and reliability of capacitors for use in power electronics. These tests include:

Routine tests

- a) Sealing test (may not be required if there is no liquid material)

- b) External inspection
- c) Voltage test between terminals
- d) Voltage test between terminals and case
- e) Capacitance and $\tan \delta$ measurements
- f) Test of internal discharge device

Type tests

- a) Mechanical tests (as per IS/IEC 60068 series)
Tests for terminals for their robustness – tensile strength, flexural strength, torsion resistance, torque resistance and solderability- and Vibration and Shock test
- b) Voltage test between terminals
- c) Voltage test between terminals and case
- d) Surge discharge test
Charging through a DC source and short circuiting 5 times. There shall be no failure and capacitance shall not vary more than a specified limit.
- e) Self-healing test (only for self-healing type capacitors)
- f) Environmental testing
Tests for temperature change, and Damp heat in accordance with IS/IEC 60068 series standards for Environmental Testing.
- g) Measurement of the tangent of the loss angle ($\tan \delta$)
- h) Thermal stability test
- i) Test of internal discharge device
- j) Resonance frequency measurement
Self inductance is measured from the resonance frequency which shall be below specified limit.
- k) Endurance test between terminals
- l) Disconnection test on fuses
- m) Destruction test
- d) Safety Requirements:** The standard includes safety considerations to mitigate risks associated with the operation and maintenance of series capacitors. This involve guidelines for installation and operation, discharge resistors, case connections and protection of environment.

5.6 ac Motor Capacitors

IS 2993: AC MOTOR CAPACITORS

Scope: This standard is applicable to motor capacitors intended for connection to windings of asynchronous motors supplied from a single-phase system having a frequency up to and including 100 Hz, and to capacitors to be connected to three-phase asynchronous motors so that these motors may be supplied from a single-phase system.

a) Specifications:

- ❖ Based on the minimum total life as per design, capacitors as designated as
 - o Class A – 30000 h
 - o Class B – 10000 h
 - o Class C – 3000 h
 - o Class D – 1000 h
- ❖ Class of Safety Protection
 - o P2 – Designed to fail in the open circuit mode only and is protected against fire and shock hazard
 - o P1 – May fail in the open circuit or short circuit mode, and is protected against fire or shock hazard.
- ❖ Class of Safety Protection
 - o P2 – Designed to fail in the open circuit mode only and is protected against fire and shock hazard
 - o P1 – May fail in the open circuit or short circuit mode, and is protected against fire or shock hazard.
 - o P0 – No specific failure protection
- ❖ Operating Temperature
 - o Minimum -40 °C, -25 °C, -10 °C or 0 °C
 - o Maximum 55 °C, 70 °C, 85 °C, or 100 °C.
- ❖ Damp Heat Severity
 - o Between 4 days and 56 days. Preferred 21 days.
- ❖ Climatic Category has to be marked on the capacitor in the fashion, for example, 25/85/21 which indicate minimum temperature -25 °C, maximum temperature 85 °C, and damp heat severity of 21 days.

b) Safety Considerations:

- ❖ **Insulation:** Ensures adequate insulation to withstand voltage stresses and prevent electrical breakdown.

- ❖ **Overpressure Protection:** Incorporates measures to prevent the buildup of excessive internal pressure, reducing the risk of capacitor failure or rupture.
- ❖ **Discharge Resistor:** Includes a discharge resistor to safely discharge the capacitor when not in use, reducing the risk of electrical shock during maintenance.
- ❖ **Environmental Compatibility:** Ensures the capacitor's materials and construction are suitable for the intended operating environment, considering factors like temperature, humidity, and vibration.

c) Testing Requirements:

- ❖ Routine tests
 - a. Sealing test, if applicable
 - b. Voltage test between terminals
 - c. Voltage test between terminals and case
 - d. Visual examination
 - e. Capacitance measurement
 - f. Tangent of loss angle
- ❖ Type tests
 - a. Visual examination, Check markings, Check of dimensions
 - b. Mechanical tests (as per IS/IEC 60068 series)
 - ❖ Robustness of terminations – Tensile, Bending, Torsion, Torque
 - ❖ Soldering
 - ❖ Vibration
 - c. Sealing tests (if applicable)
 - d. Endurance test
 - e. Damp heat test (as per IS/IEC 60068 series)
 - f. Voltage test between terminals
 - g. Voltage test between terminals and case
 - h. Self-healing test (if applicable)
 - i. Destruction test (for Classes of safety P1 and P2 type only)

d) The standard also specifies

- ❖ Permissible overloads
- Safety requirements

Marking requirements

Guide for installation and operation

Environmental Compatibility: Ensures the capacitor's materials and construction are suitable for the intended operating environment, considering factors like temperature, humidity, and vibration.

5.7. AUTOMATIC POWER FACTOR CORRECTION (APFC) PANELS

Capacitor banks with associated switching devices and control and protective circuits form automatic power factor correction panels and are widely used for dynamic power factor correction at distribution point. APFCs enable us to maintain power factor at the desired level during varying load conditions, avoid over compensation during low load conditions and help to avoid penalties imposed by electric supply companies for low power factor.

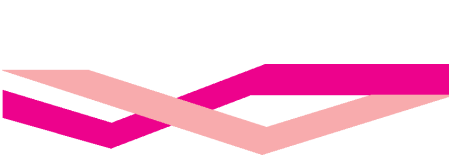
Automatic Power Factor Controller is defined as per standard as “Device designed to sense power factor of the load connected to the power line and to control the switching on and off of the steps of the capacitor bank, in order to correct the power factor to the target value”.

Requirements of APFC – IS 16636:2017

- a) Verification of Output - The output reactive power kVAR of the panel and of each individual stage shall be equal to the declared value within the tolerance limit specified in the standard
- b) Dielectric tests – Ability to withstand temporary over voltages and transient over voltages
 - i) Insulation resistance test
 - ii) Power frequency withstand voltage test
 - iii) Voltage withstand test between phases
 - iv) Lightning Impulse voltage withstand test – A standard lightning impulse of 1.2/50 μ s of specified peak voltage by an impulse voltage generator used for the test
- c) Temperature rise test
- d) Verification of the effectiveness of protective circuit – Resistance between exposed conductive part and incoming protective conductor shall not exceed 0.1 Ω
- e) Short circuit withstand test – based on the declared short circuit current capacity of the panel
- f) Verification of clearances
- g) Verification of mechanical operation

- h) Verification of the degree of protection of enclosure – Based on the declared degree of protection (IP code) as per IS/IEC 60529
- i) Verification of the effectiveness of automatic PF correction
- j) Measurement of transient over currents due to capacitor switching – the transient over current shall not exceed $10 I_N$ for electronically switched panels and $100 I_N$ for others.

The standard also gives guidelines for design, installation, operation and safety of APFC Panels.



CHAPTER VI

TESTING REQUIREMENTS FOR POWER CAPACITORS

CHAPTER VI

TESTING REQUIREMENTS FOR POWER CAPACITORS

Testing power capacitors is essential to ensure their proper functioning and reliability in electrical systems. Major tests for testing power capacitors are:

❖ **Visual Inspection:**

- Check for any physical damage, such as cracks, bulges, or leaks.
- Ensure proper connections and terminal integrity, Connection diagram if required.
- Marking details as required by the standard

❖ **Electrical Performance Tests:**

- *Capacitance Measurement:* Verify the capacitance value of the capacitor to ensure it meets specified requirements.
- *Dissipation Factor (or Loss Angle):* Measure the losses within the capacitor when subjected to an AC voltage, indicating its efficiency and performance.

❖ **Safety Tests:**

- *Lightning Impulse Voltage Test:* Evaluate the capacitor's ability to withstand transient overvoltage conditions without failure, ensuring its resilience to voltage spikes.
- *Discharge Test:* To verify the capacitor's ability to discharge its stored energy within a specified time period.
- *High Voltage withstand test:* To evaluate the capacitors ability to withstand short time high voltages without failure.

❖ **Environmental Tests:** This include tests to evaluate the capacitor's performance under various environmental conditions such as temperature, humidity, and vibration. There are a series of standards under IS/IEC 60068 covering various environmental tests in detail.

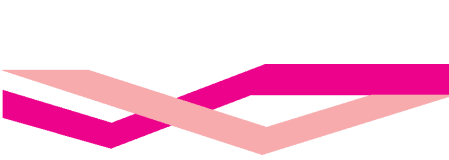
❖ **Thermal Stability Test:** To assess the capacitor's performance under temperature variations and ensure it operates within specified limits without degradation.

❖ **Endurance Test:** Subject the capacitor to prolonged operation under specified conditions to assess its long-term reliability and durability.

❖ **Destruction Test:** To assess the behaviour of the capacitor during failure.

6.1 Major laboratories in India for Capacitor testing:

- **Central Power Research Institute (CPRI) Bangalore**
- **Electrical Research and Development Association, Vadodara**
- **Hi Physics Laboratory India Pvt Ltd, Pune.**



CHAPTER VII

CAPACITOR SELECTION

CHAPTER VII

CAPACITOR SELECTION

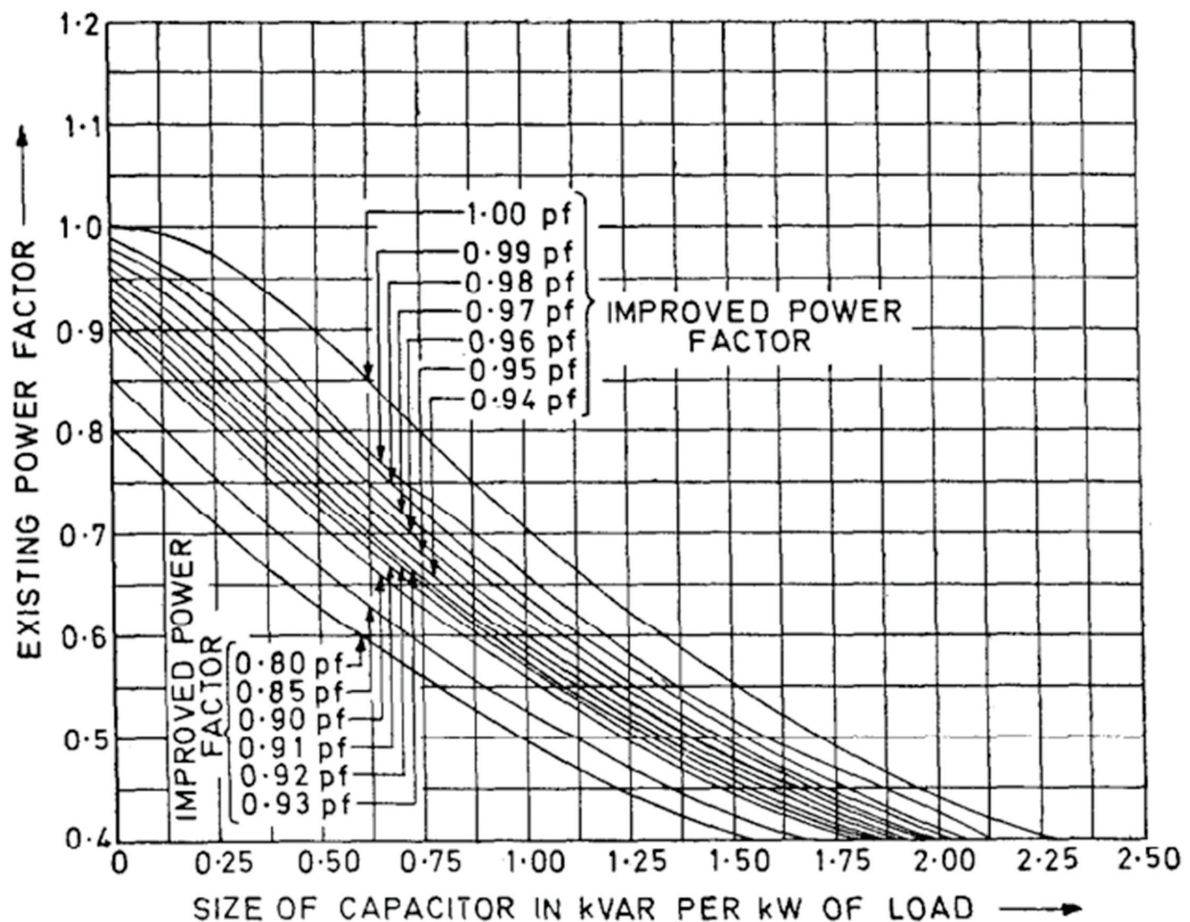
Power capacitor selection and application guidelines provide a systematic approach to choosing and utilizing capacitors effectively in electrical systems, particularly for tasks such as power factor correction and voltage regulation. These guidelines help ensure that the selected capacitors meet the specific requirements of the application, operate efficiently, and comply with relevant standards and regulations.

7.1 Factors Influencing Capacitor Selection: Power capacitor selection is influenced by various factors that need to be considered to ensure optimal performance, reliability, and efficiency in electrical systems. Some key factors influencing power capacitor selection:

- ❖ **Power Factor Correction Requirement:** Power factor correction (PFC) is the process of improving the power factor of an electrical system by adding power capacitors to counteract the effects of reactive power generated by inductive loads. The required level of power factor correction, such as achieving unity power factor or a specific target, influences the selection of capacitors.
- ❖ **Reactive Power Demand:** The reactive power (VAR) demand of the electrical system determines the capacitance required for power factor correction. Capacitors are selected to provide the necessary reactive power compensation to improve the power factor.
- ❖ **System Voltage:** Capacitors must be rated to withstand the system voltage without breakdown. The voltage rating of capacitors should match or exceed the maximum voltage of the electrical system.
- ❖ **Harmonic Content:** Harmonic distortion in the electrical system can affect capacitor performance and reliability. Capacitors with harmonic filtering capabilities may be necessary to mitigate harmonic issues and prevent damage.
- ❖ **Dielectric Material:** The dielectric material used in power capacitors plays a crucial role in their performance, reliability, and suitability for specific applications in electrical systems. Different dielectric materials offer varying characteristics in terms of dielectric strength, temperature stability, energy storage capacity, self-healing properties, and cost.
- ❖ **Equivalent Series Resistance (ESR):** Equivalent Series Resistance (ESR) is a critical parameter in power capacitors, especially in applications where high efficiency, low losses, and high frequency performance are essential. ESR represents the internal resistance of a capacitor, which includes the resistance of the dielectric material, the internal connections, and any other parasitic resistances within the capacitor construction. In power capacitors, ESR can significantly affect the performance and efficiency of the capacitor and the overall system.

- ❖ **Environmental Conditions:** Environmental factors such as temperature, humidity, and altitude can affect capacitor performance and lifespan. Capacitors should be selected with ratings suitable for the environmental conditions of the installation site.
- ❖ **Compliance and Standards:**
In India, power capacitors and their usage are subject to various compliance requirements and standards to ensure their safety, performance, and reliability. These need to be factored during their selection and usage.
- ❖ **Economics:** Cost-effectiveness is an important factor in capacitor selection, balancing performance requirements with budget constraints. It's essential to evaluate the total cost of ownership, considering factors such as initial purchase cost, reliability, maintenance, and system downtime associated with capacitor failure.

7.2 IS 7752 (Part 1) : 1975 gives a detailed guidance for selection of capacitors for power factor correction including calculations and recommended ratings of capacitors.



(Recommended size of capacitor for desired power factor correction as given IS 7752 (Part 1))

CHAPTER VIII
NATIONAL ELECTRICAL CODE
OF INDIA- 2023

CHAPTER VIII

NATIONAL ELECTRICAL CODE OF INDIA- 2023 CODE OF INDIA- 2023

The **National Electrical Code of India (NEC)** is an all-inclusive guide and code of practice for Electrical Installations in the country, prepared by BIS, providing guidelines for regulating electrical installation practises across the country.

The code contains:

- Good practices for the selection of electrical equipment forming part of an electrical installation;
- Recommendations concerning safety in the wiring of electrical installations,
- General safety procedures and practices in electrical work; and
- Additional precautions that must be taken when using electrical equipment in hazardous environments such as an explosive or active atmosphere.

The NEC contains 8 Parts and 49 sections.

Part1	General and Common Aspects (22 Sections)
Part2	Electrical installations in stand by generating stations and captive substations (2 Sections)
Part3	Requirements for special installations or locations (22 Sections)
Part4	Electrical installations in industrial buildings
Part5	Outdoor installations (3 Sections)
Part6	Electrical installations in agricultural premises
Part7	Electrical installations in Hazardous area
Part8	Solar Photovoltaic (PV) power supply systems

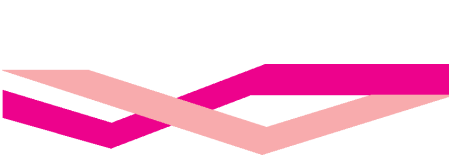
Electrical accidents will be drastically reduced if the electrical engineers design the electric installation considering the requirements mentioned in the National Electrical Code of India, using the right protection devices for the right equipment, and if electricians and other professionals carry out proper verification (initial and periodic) in accordance with the criteria given in the NEC in adherence to the safety regulations.

8.1 Power Capacitors in NEC

NEC has devoted an entire section (Part 1/Section 20.1 Power Quality Improvement – pf Correction and harmonics control) for power quality improvement in which usage of power capacitors are discussed. The code discusses causes of low power factor, disadvantages of it, economics of power factor improvement and use of capacitors and APFCs for the same. It also gives method of selection, methods of connection and calculations of ratings of capacitor required for power factor correction.



Section 21 Energy Efficiency Aspect discuss the need of power factor correction and the requirement of maintaining minimum power factor of 0.95 for industries and 0.90 for medium and large commercial buildings. The need of active harmonic filters are also discussed when the Total Harmonic Distortion (THD) exceeds the maximum specified value of 5% for commercial and industrial establishments.



CHAPTER IX

REGULATORY REQUIREMENTS

CHAPTER IX

REGULATORY REQUIREMENTS

Standards are basically voluntary in nature, adopted by the manufacturer to ensure quality of their product. They can also form a part of contract between purchaser and manufacturer and become mandatory in a sense.

However, the government may enforce conformity to standard by including them in their regulatory framework, considering safety, health, prevention of unfair trade practices or national interest. In India, Quality Control Orders (QCOs) are issued by various departments of Government of India making conformity to Indian Standard and BIS Certification mandatory for certain products.

The Electrical Capacitors (Quality Control) Order, 2017 has made the following capacitors mandatory to conform to given Indian Standards.

1.	IS 2993	A.C. motor capacitors dated 01st	QCO - SO 2434(E), August 2017
2.	IS 13340	Power Capacitors of Self-healing Type for AC Power Systems having Rated Voltage up to 1000 V	Date of Implementation 01/02/2018
3.	IS 13585 Part 1	Shunt Power Capacitors of the Non-Self-Healing Type for AC System having a Rated Voltage up to and Including 1000V Part-1 General Performance, Testing and Rating Safety Requirements Guide for Installation and Operation	

According to this QCO, no person shall manufacture, import, store or store for sale, sell or distribute capacitors as given in the table which do not conform to the specified Indian Standard and which do not bear the standard mark of BIS. Foreign manufacturers also need to take BIS Certification for exporting to India. Bureau of Indian Standards Act 2016 specifies fines and imprisonment for violation of QCOs.

- *Central Electricity Authority (CEA)*: CEA is a statutory organization responsible for formulating policies and regulations related to the power sector in India.
- *Indian Electricity Act 2003*: Consolidated laws related to generation, transmission, distribution, trading and use of electricity.
- Central Electricity Authority (Measures relating to Safety and Electric Supply) Regulations, 2023 mentions that **the relevant standards including National Electrical Code and National Building Code shall be followed to carry out the purpose of these regulations**, with reference to construction, installation, protection, operation and maintenance of electric supply lines and apparatus. International standards shall be used only when Indian Standards are not available.



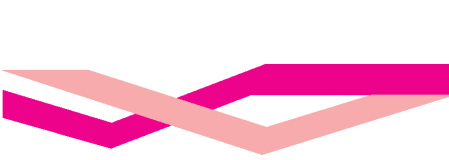
CHAPTER X

REFERENCES:

CHAPTER X

REFERENCES:

- “Introduction to Electric Circuits” by James A. Svoboda and Richard C. Dorf
- “Power System Capacitors” by Ramasamy Natarajan
- Charles K. Alexander and Matthew N.O. Sadiku. “Fundamentals of Electric Circuits.” McGraw-Hill Education, 2013.
- “Electric Circuits” by Nilsson and Riedel
- “Power Capacitors: Theory and Applications” by P.S. Bimbhra
- “Power Capacitors: Testing and Failure Analysis” by G. R. Jones and D. H. W. Davies
- **Bureau of Indian Standards (BIS- <https://www.bis.gov.in/>) publications.**



ANNEXURE

Annex 1 Indian standards on Power Capacitors

IS Number	IS Title	Aspect	Degree of Equivalence with IEC Standard
IS 13585 (Part 1) : 2012 / IEC 60931-1 : 1996	Shunt power capacitors of the non - Self - Healing type for ac systems having a rated voltage up to and including 1 000 v: Part 1 general - Performance, testing and rating - Safety requirements - Guide for installation and operation (First Revision)	Product Specification	Identical under dual numbering
IS 13585 (Part 2) : 2013 / IEC 60931-2: 1995	Shunt power capacitors of the non - Self - Healing type for A.C. systems having a rated voltage up to and including 1 000 v: Part 2 ageing test, self-healing test and destruction test	Methods of tests	Identical under dual numbering
IS 13585 (Part 3) : 2013 / IEC 60931_3: 1996	Shunt power capacitors of the non - Self - Healing type for A.C. systems having a rated voltage up to and including 1 000 V: Part 3 internal fuses	Product Specification	Identical under dual numbering
IS 13340 (Part 1): 2012 / IEC 60831 - 1 : 2002	Shunt power capacitors of the self-Healing type for AC systems having a rated voltage up to and including 1000 V: Part 1 general - Performance, testing and rating - Safety requirements - Guide for installation and operation (First Revision)	Product Specification	Identical under dual numbering
IS 13340 (Part 2): 2012 / IEC 60831_2 : 1995	Shunt power capacitors of the self-Healing type for AC systems having a rated voltage up to and including 1000 V: Part 2 ageing test, self - Healing test and destruction test	Methods of tests	Identical under dual numbering
IS 13925 (Part 1): 2012 / IEC 60871-1 : 2005	Shunt capacitors for AC power systems having a rated voltage above 1000 V: Part 1 general (First Revision)	Product Specification	Identical under dual numbering
IS 13925 (Part 2): 2002 / IEC	Shunt capacitors for A.C. power systems having a rated voltage	Methods of tests	Identical under dual numbering

IS Number	IS Title	Aspect	Degree of Equivalence with IEC Standard
60871-2	above 1000 V: Part 2 endurance testing (First Revision)		
IS 13925 (Part 3): 2002 / IEC 60871-3	Shunt capacitors for A.C. power systems having a rated voltage above 1000 V: Part 3 protection of shunt capacitors and shunt capacitor banks	Others	Identical under dual numbering
IS 13925 (Part 4): 2013 / IEC 60871_4: 1996	Shunt capacitors for A.C. power systems having a rated voltage above 1000 V: Part 4 internal fuses	Product Specification	Identical under dual numbering
IS 11530 : 1985	Specification for voltage grading capacitor	Product Specification	Indigenous
IS 11548 : 1986	Specification for capacitors for surge protection for use in voltage system above 650 V and up to 33 KV	Product Specification	Indigenous
IS 13203 : 1991 (IEC 60996)	Method for verifying accuracy of tan delta measurements	Methods of tests	Modified/ Technically Equivalent
IS 13666 : 1993	Energy storage capacitors - Specification	Product Specification	Indigenous
IS 15395 : 2003 / IEC 61642	Industrial AC networks affected by harmonics application of filters and shunt capacitors	Others	Identical under dual numbering
IS 16636 : 2017	Automatic power factor correction (APFC) panels for voltage rating up to and including 1000 V	Product Specification	Indigenous
IS/IEC 60143-1 : 2015	SERIES CAPACITORS FOR POWER SYSTEMS PART 1 : GENERAL First Revision	Product Specification	Identical under single numbering
IS 9835 (Part 2) : 2021 / IEC 60143-2:2012	SERIES CAPACITORS FOR POWER SYSTEMS PART 2: PROTECTIVE EQUIPMENT FOR SERIES CAPACITOR BANKS	Product Specification	Identical under single numbering
IS/IEC 60143-3 : 2015	SERIES CAPACITORS FOR POWER SYSTEMS PART 3: INTERNAL FUSES First Revision	Product Specification	Identical under single numbering
IS/IEC 60143-4 :	SERIES CAPACITORS FOR	Product	Identical under

IS Number	IS Title	Aspect	Degree of Equivalence with IEC Standard
2010	POWER SYSTEMS PART 4: THYRISTOR CONTROLLED SERIES CAPACITORS	Specification	single numbering
IS/IEC 60110-1 : 1998	Power Capacitors for Induction Heating Installations Part 1 General (First Revision)	Product Specification	Identical under single numbering
IS/IEC/TS 601 10-2 : 2000	POWER CAPACITORS FOR INDUCTION HEATING INSTALLATIONS PART 2: AGEING TEST DESTRUCTION TEST AND REQUIREMENTS FOR DISCONNECTING INTERNAL FUSES	Methods of tests	Identical under single numbering
IS/IEC 61071 : 2017	CAPACITORS FOR POWER ELECTRONICS	Product Specification	Identical under single numbering
IS/IEC 60358-1 : 2012	COUPLING CAPACITORS AND CAPACITOR DIVIDERS PART 1 GENERAL RULES	Safety Standard	Identical under single numbering
IS/IEC 60358-2 : 2013	COUPLING CAPACITORS AND CAPACITOR DIVIDERS PART 2 AC OR DC SINGLE-PHASE COUPLING CAPACITOR CONNECTED BETWEEN LINE AND GROUND FOR POWER LINE CARRIER-FREQUENCY PLC APPLICATION	Methods of tests	Identical under single numbering
IS 11298 (Part 1): 1985IEC 60674-1	Specification for plastic films for electrical purposes: Part 1 definitions and general requirements	Product Specification	Identical under dual numbering
IS 11298 (Part 2): 2024IEC 60674-2: 2016	Plastic Films for Electrical Purposes - Specification Part 2 Methods of Test (Second Revision) Methods of tests		Identical under dual numbering
IS 11298 (Part 3/Sec 1) : 1991	Plastic films for electrical purposes - Specification: Part 3 specifications for individual materials: Sec 1 polypropylene films for capacitors (First Revision)	Product Specification	Modified/ Technically Equivalent

IS Number	IS Title	Aspect	Degree of Equivalence with IEC Standard
IS 11298 (Part 3/ Sec 2) : 1990	Plastic films for electrical purposes - Specification: Part 3 specifications for individual materials: Sec 2 metallized polypropylene films	Product Specification	Indigenous
IS 13067 : 1991	Impregnants for power capacitors Specification	Product Specification	Modified/ Technically Equivalent
IS 9335 (Part 3/ Sec 2) : 1985	Specification for cellulosic papers for electrical purposes: Part 3 specifications for individual materials: Sec 2 capacitor paper	Product Specification	Modified/ Technically Equivalent
IS 9335 (Part 3/ Sec 4) : 1984	Specification for cellulosic papers for electrical purposes: Part 3 specifications for individual materials: Sec 4 electrolytic capacitor paper	Product Specification	Modified/ Technically Equivalent
IS 9335 (Part 1) : 2024	Cellulosic Papers for Electrical Purposes Part 1 Definitions and General Requirements	Terminology	Modified/ Technically Equivalent
IS 9335 (Part 2) : 2024 IEC 60554-2: 2001	Cellulosic Papers for Electrical Purposes Part 2 Methods of Test (Second Revision)	Methods of tests	Identical under dual numbering
IS 1885 (Part 42):	1993 / IEC Pub 60 (436) (1990) Electrotechnical vocabulary: Part 42 Power Capacitors	terminology	Identical under dual numbering
IS 1885 (Part 45): 1977	Electrotechnical vocabulary: Part 45 capacitors	Terminology	Modified/ Technically Equivalent
IS 824 : 2021 / IEC 60063: 2015	Preferred number series for Resistors and Capacitors	Code of Practice	Identical under dual numbering
IS 9402 : 1980 (IEC 549)	Specification for high Voltage fuses for external protection of shunt power capacitors	Product Specification	Modified/ Technically Equivalent
IS 2993 : 1998 (IEC 252)	A.C. motor capacitors (Second Revision)	Product Specification	Identical under dual numbering
IS 7752 (Part 1) : 1975	Guide for improvement of power factor in consumer installations: Part 1 low and medium supply voltages	Code of Practice	Indigenous



IS Number	IS Title	Aspect	Degree of Equivalence with IEC Standard
IS 4486 : 1967	Recommended methods for the determination of the permittivity and dielectric dissipation factor of electrical insulating materials at power, audio and radio frequencies including metre wavelengths	Methods of tests	Indigenous